

RMC Mix Design Collection – M10 & M20

Synthetic mix-design records with multiple M20 formulations and deliberately varied cement contents.

M10 Concrete – Mix Design M10-A

Application: PCC, leveling and non-structural applications **Target Slump:** 75 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	220	kg/m ³	Primary cementitious binder
Fly Ash	0	kg/m ³	Supplementary cementitious material
Water	190	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	850	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	470	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	700	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	0.0	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.864	-	Illustrative design value
Target Mean Strength	16.6	MPa	Illustrative 28-day target

Design remarks: Low-strength illustrative mix with relatively high water-to-cement ratio. Cement content is intentionally different from the M20 design.

M20 Concrete – Mix Design M20-A

Application: General RCC and structural applications **Target Slump:** 85 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	300	kg/m ³	Primary cementitious binder
Fly Ash	40	kg/m ³	Supplementary cementitious material
Water	175	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	760	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	430	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	670	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	2.8	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.515	-	Illustrative design value
Target Mean Strength	26.6	MPa	Illustrative 28-day target

Design remarks: Illustrative M20 mix using blended cementitious content. Aggregate quantities and admixture dosage are sample values.

M20 Concrete – Mix Design M20-B

Application: General RCC with increased workability **Target Slump:** 100 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	330	kg/m ³	Primary cementitious binder
Fly Ash	30	kg/m ³	Supplementary cementitious material
Water	168	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	745	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	425	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	665	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	3.4	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.467	-	Illustrative design value
Target Mean Strength	26.6	MPa	Illustrative 28-day target

Design remarks: Alternative M20 design with higher cement content and lower illustrative water-to-cementitious ratio.

Dataset note: These are synthetic RMC mix-design records created for RAG ingestion, retrieval, comparison, and question-answering tests. They are not approved construction designs. Actual mix proportions must be established and validated using project requirements, material properties, laboratory trials, moisture corrections, and applicable standards.